

Austin J. Mendoza, Ph.D.

Postdoctoral Research Associate

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Education

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|-------------|---|
| 2019 - 2025 | Lehigh University , Bethlehem, PA
<i>Ph.D. in Biology, Neuroscience focus</i>
Electrical synapses and sensory attention at the thalamic reticular nucleus., Advised by Julie S. Haas, Ph.D. |
| 2017 - 2019 | University of California, Fresno , Fresno, CA
<i>M.S. in Biology, with distinction</i>
Modeling visual preception, learning, and memory of wood ants navigating in naturalistic environments., Advised by David D. Lent, Ph.D. |
| 2012 - 2017 | University of California, Fresno , Fresno, CA
<i>B.S. in Biology</i> |

Publications

1. **Mendoza. AJ** & JS Haas. (2025). Electrical coupling within thalamocortical networks cumulatively reduces cortical correlation to sensory inputs. In Submission, J. neurosci. BioRxiv. Doi:10.1101/2025.08.21.671565
2. Rolón-Martínez S, **Mendoza AJ**, Angeloni CF, Vogler NW, Drotos AC, Aizenberg M, et al. (2026) Thalamic reticular neurons provide cell type-specific modulation of sound processing in the auditory thalamus. PLoS Biol 24(3): e3003693. <https://doi.org/10.1371/journal.pbio.3003693>
3. **Mendoza. AJ** & JS Haas. (2022). Intrinsic sources and functional impacts of asymmetry at electrical synapses. eNeuro. Doi:10.1523/ENEURO.0469-21.2022

Honors & Awards

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| 2022 | Thorne Fellowship , Lehigh University, Department of Biological Sciences
Semester long department research fellowship recognizing research progress of select Ph.D. Candidates. |
| 2022 | Summer Fellowship , Lehigh University, College of Arts and Sciences
Summer research fellowship awarded to a select few graduate students in the College, nominated by department. |

Conference Presentations

1. Organization for Computational Neurosciences (2025). Electrical coupling within thalamocortical networks cumulatively reduces cortical correlation to sensory inputs. Poster.
2. Society for Neuroscience (2023). Understanding variations in MGB tone responses due to TRN synaptic connectivity. Poster.
3. Organization for Computational Neurosciences (2021). Neuronal heterogeneity underlies electrical synapse asymmetry and spike time variability in coupled neurons. Poster.
4. Society for Neuroscience (2018) Modeling visual perception, learning, and memory of wood ants navigating in naturalistic environments. Poster.